SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5

Code of Conduct Shanika (192511387)

I ~~Betty . A~~ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Betty

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1) Compare Type 1 and Type 2 Hypervisors.

Feature	Type 1 - Hypervisor	Type - 2 Hypervisor
Installation	Runs directly on hardware	Runs on top of an operating system
Performance	High	Lower due to OS overhead
Security	More secure	Less secure
Reliability	Highly Reliable	Less Reliable
Examples	VMware ESXi, Microsoft Hyper-V, Xen	VMware Workstation Oracle VirtualBox

Justification:

Type-1 Hypervisor is the better choice because it offers higher performance, better security, lower latency, and greater reliability, making it suitable for mission-critical cloud applications.

2) Virtualization architecture for workload isolation.

Hardware - assisted virtualization with a Type-1 Hypervisor.

Major attack surfaces

Hypervisor attacks

Shared storage or network attacks

Security measures:

Strong VM isolation

Virtual firewalls

Network segmentation (VLAN)

Encryption of data

Regular security patches

Role - Based Access Control (RBAC)

3) VM Host Utilization Analysis

Given :

CPU : 92 %

Memory = 88%

Storage I/O Latency = 35 ms

Average VM Boot Time = 170 s

Likely Bottlenecks:

CPU overload

High memory usage

Slow storage performance

Delayed VM startup

Corrective Actions:

Migrate VMs using live migration

Add CPU and RAM resources

Upgrade to SSD / NVMe storage

Balance workloads across hosts

Enable resource monitoring and auto-scaling.

4) How Software Defined Data Center (SDDC) improves agility.

Compute Virtualization:

Enables quick VM creation, migration, and resource allocation.

Storage Virtualization:

Combines storage resources into a single pool, improving utilization and scalability.

Network Virtualization:

Creates virtual networks, improves security through micro-segmentation, and allows rapid network configuration.

Conclusion:

SDDC improves agility by automating compute, storage, and networking, reducing deployment time, increasing scalability, and simplifying management.

5) Secure Identity and Privacy Design for Multi-cloud Federation.

Identity mismatches between cloud providers cause authentication and access problems.

Solution:

Use Federated identity management

Implement single sign-on (SSO)

Use OAuth 2.0, SAML, or OpenID connect

Enable Multi-Factor Authentication

Apply Role-Based Access Control

Encrypt user data and maintain audit logs.

Result :

Federated identity provides secure authentication , protects user privacy . reduces duplicate accounts , and enables seamless access across multiple cloud providers.